

Aditya Bhushan Singh

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Education

University of Cincinnati, Carl H. Lindner College of Business, Cincinnati, OH

Master of Science in Business Analytics | GPA: 3.9 | July 2017

BMS Institute of Technology, Visvesvaraya Technological University, Bangalore, India

Bachelor of Engineering in Electrical and Electronics | GPA: 79% | June 2012

Professional Experience

Wiser Solutions Inc., Toronto, Ontario

Director of AI/ML Platform | May 2025 -

- Leading two cross-functional AI/ML and Analytics engineering teams (8+ engineers) to build and scale production ML & LLM platforms that drive profitability and customer retention, aligning model build and deployment, MLOps strategy & data infrastructure with product, operations, and GTM priorities.
- Define and drive enterprise AI adoption strategy as executive steering committee member, leading platform standardization initiatives spanning tool evaluation, rapid prototyping infrastructure, organization-wide adoption programs, and vendor licensing negotiations.
- Rebuilt the core internal matching platform from the ground up, processing 10MM+ records daily across 100+ customers at <3¢ per match, powering all customer-facing Wiser products. Built a vector embedding training service on EKS to scale embedding generation for 100+ customer catalogs using an event-driven architecture.
- Engineered ML infrastructure for NLP, vector embeddings, LLMs, and agentic AI systems with scalable MLOps and LLMOps frameworks for enterprise-wide model training, deployment, and orchestration.
- Built scalable MLOps infrastructure for training and deploying Sentence Transformer models in PyTorch on SageMaker GPU clusters with CUDA acceleration, memory-efficient batching, and sharded pipelines which reduced training time by 67% and embedding costs to <\$7 per 1M records. These engines now power all customer-facing Wiser products.
- Built reusable NLP platform infrastructure supporting named entity recognition (NER), information extraction, and information retrieval workflows using BM25, with standardized data handling (Pandas/NumPy) and cloud-native architectures, reducing time-to-market for new NLP features by 30% across customer-facing products.
- Building an AI/BI conversational analytics platform with a semantic layer, metadata knowledge graph, caching, and optimized query pipelines, enabling natural language access to internal databases with retrieval times <15 secs. Early adoption improved customer retention by 12%, accelerated sales-qualified leads, cut internal data analysis time by ~30%, and reduced support costs by \$200K+ per year.
- Engineered autonomous agentic ML systems using CrewAI framework with full MLOps orchestration on SageMaker, including automated model training, deployment pipelines, logging/alerting, and autoscaling for intelligent proxy management thereby eliminating manual intervention and unlocking \$4M+ annual operational savings.

Loblaw Companies Limited, Toronto, Ontario

Director of Data Science | Jun 2022 - May 2025

Data & Machine Learning Platforms

- Built a cross-functional team (engineering, data science, front-end, UX, QA, contractors) that delivered \$50M+ in incremental annual revenue for two consecutive years by building and scaling ML platform supporting demand forecasting and price optimization models.
- Built and scaled Loblaw's first cloud-native promotional optimization platform, deploying thousands of ML models weekly across 200+ store groups, powering 100K+ optimized recommendations per week.
- Architected a microservices-based pricing platform using REST APIs for sync communication and Google Pub/Sub for event-driven workloads. Designed for scalability and extensibility across pricing channels.
- Developed and deployed ensemble-based ML models for shelf, promo, and markdown pricing. Orchestrated via Airflow and Docker, deployed to Kubernetes these models reduced pricing recommendation volumes by 20%, easing network load and improving decision quality.
- Integrated a Google Gemini-powered chatbot into the pricing app, built on RAG architecture using LangChain and Text2SQL. Enables real-time stakeholder queries with semantic and hybrid search over GCP-hosted datasets.
- Led the design and implementation of ML solutions supporting environmental sustainability, including energy optimization models for HVAC, refrigeration, and condensers which drove \$30M+ in annual energy savings.
- Formed and led an AI/ML leadership team to shape strategy around modeling standards, data governance, and platform architecture which helped in reducing infra sprawl and aligning ML practices across teams.

Kinaxis Inc., Toronto, Ontario

Sr. Business Data Scientist | Feb 2021 – Jun 2022

CPG Demand Forecasting & Sensing

- Practice Lead for the CPG advanced analytics team building Demand Forecasting & Sensing solutions for multiple

Fortune 500 clients leveraging AI/ML models such as lightGBM, xgboost, etc. to improve short- & long-term demand forecasting strategy

- Led a team of 6 Data Scientists & Engineers to build a scalable E2E Machine Learning product to integrate with Rapid Response which will introduce Demand Sensing capabilities to long term forecasting algorithms
- Improved Forecast Accuracy & Bias by 25 & 40 percentage points respectively using lightGBM forecasting across CPG organizations leading to operational savings of \$8MM & reduced Demand Planner hours by ~40% over a period of 6 months

Target Corporation, Minneapolis, Minnesota

Lead Data Scientist | Sep 2019 – Feb 2021

Digital Operations

- Led a team of 7 Data Scientists working on solving ML/AI problems across Target's E-Commerce business including Channel Attribution, Guest Visit Classification, Item Taxonomy, etc. My teams work on these problems led to incremental conversion of 30% and attributable demand of \$18MM.
- Increased digital net demand by 2% MoM while working with Marketing by building a scalable xgboost model to quantify the effects of paid search and affiliate marketing dollars on the digital bottom line.
- Developed the logic & ETL pipeline to measure the performance of Target's recommendation algorithms across Web & App. Created dashboards for C-Level Executives & Business to study CTR, Conversion & Demand

Senior Data Scientist | Aug 2017 – Sep 2019

Datalyst – Supply Chain Performance Products

- Created a new product for productivity calculations for the supply chain, based on event level data. Worked on all aspects of product development including story boarding, ETL, data modeling & product analytics.
- Productivity metrics are used to forecast staffing plans, refine learning curves for new hires, perform key driver analysis on 100+ network performance attributes, etc. leading to quarterly savings of ~\$1.5MM.
- Created an in-house, reusable optimization engine using Gurobi & Python to optimize labor requirements across the supply chain, saving Target ~\$0.10 in labor cost per unit using an MIP model.

Target Corporation, Bangalore, India

Senior Business Analyst | August 2014 – July 2016

Apparels & Accessories Demand Forecasting

Achieved savings of \$15MM in annual revenues by solving the issue of improper inventory allocation by developing a tree-based model to accurately predict lost sales and identify true demand. The model had a demand estimation accuracy of ~87%

Healthcare Guest Analytics

- Created RFV segments for Target pharmacy guests using K-means clustering in R and Tableau to help identify guest characteristics and determine probability of guest churn to devise marketing programs to improve retention.

Mu Sigma Business Solutions Pvt. Ltd., Bangalore, India

Senior Business Analyst | February 2013 – August 2014

Specialty Pharmacy Providers

- Led a team of three analysts to help design and roll out the business model for SPP to improve profitability thereby helping the client save ~\$4M in rebates to be given to pharmacies.

Faculty Experience

University Canada West, Vancouver, BC

Sessional Faculty | Aug 2022 -

BUSI-650 – Introduction to Business Analytics

- Taught introductory courses for Graduate MBA students to introduce Business Analytics 101 concepts
- Course contains topics such as Statistics 101, Hypothesis Testing, Linear & Logistic Regression, Time Series Modeling, Visualizations with Tableau, Introduction to SQL
- Responsible for design, delivery & administration of course to multiple cohorts of ~40 students every semester

Great Learning

Data Science Instructor | Sep 2021 -

Business Analytics & Data Science

- I facilitate mentored learning sessions for experienced professionals. The course is offered in partnership with UT Austin's McCombs school of business